

CE 351: Design of Reinforced Concrete Structures

Course Code	CE 351
Course Name	Design of Reinforced Concrete Structures
L-T-P-C	2-1-0-3
Prerequisites	CE 301 - Strength of Materials and Structures
Intended for	UG
Distribution	Discipline Core
Approval	9th Senate (May 2016)

Course Contents

- **Design of beams**
Singly and doubly reinforced. [6 Lectures]
- **Design of columns**
Short and slender columns. [6 Lectures]
- **Design of slabs**
One way, two way and flat slabs. [6 Lectures]
- **Module IV**
Design for bond and anchorage of reinforcement. [6 Lectures]
- **Module V** Reinforcement detailing at beam-column and column-foundation junctions. [6 Lectures]
- **Module VI**
Design of Staircase, Isolated and combined footings. [6 Lectures]
- **Introduction to pre-stressed concrete design**
Basics of seismic analysis and detailing. [6 Lectures]

Text Books

1. Pillai, S.U. and Menon, D., **Reinforced concrete design**, Tata MacGraw Hill, 2009.
2. Subramanian, N, **Design of reinforced concrete structures**, xford University Press, 2014.
3. Sinha, S.N., **Reinforced concrete design**, Tata McGraw-Hill, 2002.
4. Raju, N.K. and Pranesh, R.N., **Reinforced concrete design: principles and practice**, New Age International, 2007.
5. Raju, N.K., **Prestressed concrete**, Tata MacGraw Hill, 2006.

References

1. Nawy, E.G., **Reinforced concrete: a fundamental approach**, Prentice Hall, 2009.
2. Wright, J.K. and MacGregor, J.G., **Reinforced concrete: mechanics and design**, Prentice Hall, 2015.
3. Hewson, N.R., **Prestressed concrete bridges: design and construction**, Thomas Telford, 2003.
4. Varghese, P.C., **Advanced reinforced concrete design**, Prentice Hall of India, 2009.